CS5707 Artificial Intelligence
Lab 1. Google Colab and Python

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We will use the Google Colaboratory (or Google Colab or simply Colab) in the labs of this module, which is a free Jupyter notebook environment hosted by the Google cloud services. The Colab uses the same format of Jupyter Notebook, a popular interactive development platform for Julia, Python and R.

Aims and Objectives

This lab is aimed to make you become familiar with the Google Colab environment and warm up for the Python programming language. After this session, you should achieve the following objectives:

1) To be familiar with the Colab programming environment.
2) To build a simple “Hello World” Python program with Colab, and to produce text and graphical outputs.
3) To understand the format of lab reports.
4) Python warm-up (or learning from start).

Time Required

2 hours (with supervision).

Please note that the above number is only indicative and should be regarded as an average amount of time for all students. You may require significantly more or less time depending on personal circumstances, e.g. knowledge and understanding to the relevant subject topics, experience of the software used, and level of programming skills. Please do prepare to use more time outside the officially arranged sessions.

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Task 1.1 The “Hello World” Program

Go to the Google Colab by following the link below:
https://colab.research.google.com/

You are required to sign in with your Google account if you have not done so. Just click on the “Sign In” at the top right of the screen. Or if you have already signed in, it would prompt the following screen to you, where you may load your recent notebooks, or those stored in your Google Drive or Github.

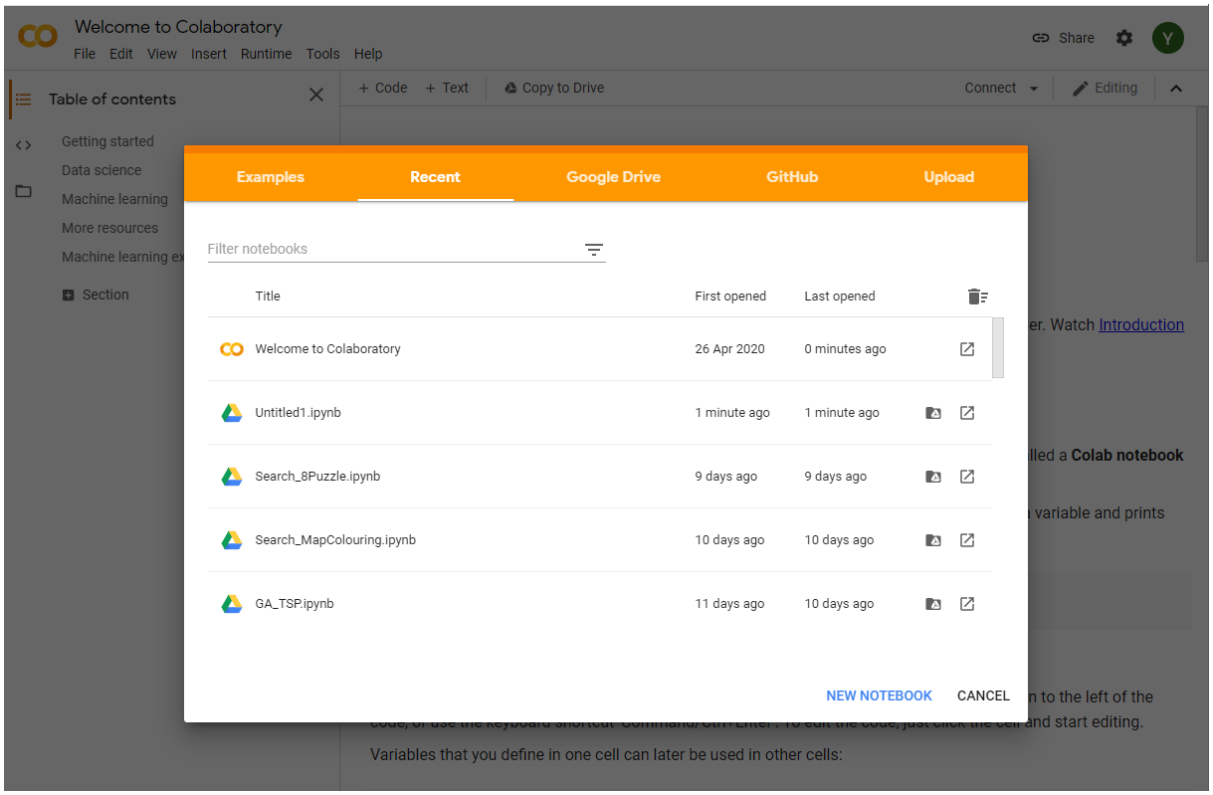


Fig 1.1 Google Colab programming environment

1.1.1 Your first Python program

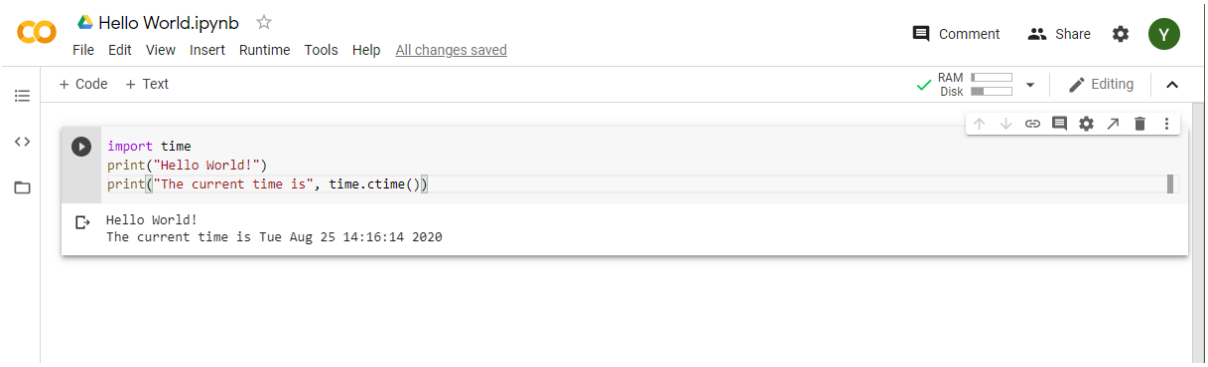
If you want to create a new notebook, simply click on the blue “NEW NOTEBOOK” at the bottom of the pop up window, or select “New notebook” from the File menu.

At the left top of the screen, you will see a default name like “Untitled#.ipynb” is given for the newly created notebook. Rename it with something meaningful, such as “Hello World.ipynb”.

Enter the following code into the blank cell, then click on menu “Runtime – Run the focused cell” (or simply hit Ctrl-Return).

```
import time
print("Hello World!")
print("The current time is", time.ctime())
```

Then you will see the running result immediately below the code that shows a message of “Hello World” and the current date/time. This is your very first Python program with Google Colab!



1.1.2 Graphical output

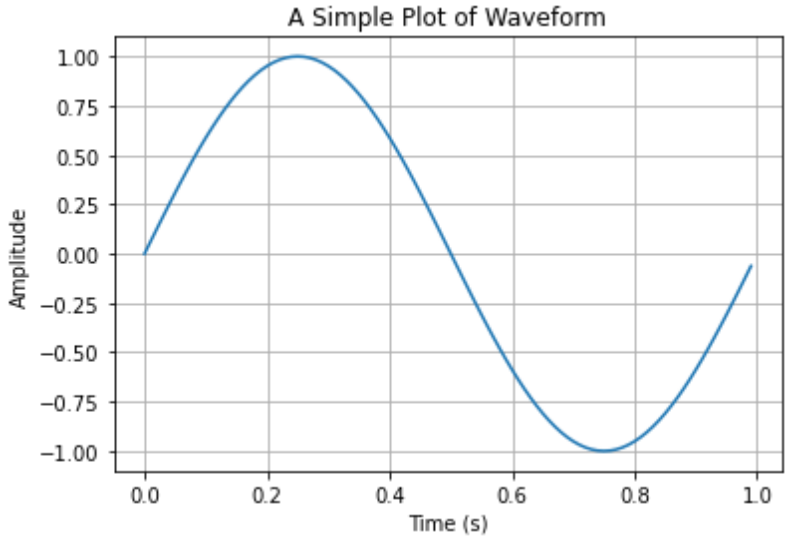
Let us add another Code cell with the following code to generate some graphical output – a simple plot of waveform:

```
import matplotlib.pyplot as plt
import numpy as np

t = np.arange(0.0, 1.0, 0.01)
s = np.sin(2 * np.pi * t)

plt.plot(t,s)
plt.title('A Simple Plot of Waveform')
plt.xlabel('Time (s)')
plt.ylabel('Amplitude')
plt.grid()
```

Run the cell, and you will have a plot is shown:



1.1.3 Code cells and Text cells

In the Colab, you have two types of cells, the **Code** cells and the Text cells. The previous code we completed was within a **Code** cell. Of course, you can still use the standard Python comments to add to your code in order to increase its readability, which you are highly encouraged to do so. However, the **Text** cells in Colab offer a richer and more friendly format where you can edit your text with any style you like just like in Google Docs or Microsoft Word, and you can even add mathematical formulas and graphics.

Let us add one **Text** cell at the very beginning of the notebook, and add the following text into the cell. Use the up/down arrows (normally appeared at the top-right corner of the cell) to adjust the position of cell.

```
# Lab 1. Introduction to Artificial Intelligence
## Problem Descriptions
Describe the problem here...
## Implementation and Results
```

The lines with '#' and '##' will create different levels of headings. Then we add another Text cell at the end of the notebook with the following text:

```
## Discussions
In this task, I have created a "Hello World" Pytyon program using Google Co
lab.
More discussions...
```

Now your notebook should look like in Fig 1.2 lab report format.

1.1.4 Format of lab reports

Please use the notebook you have just created as the template for all your future lab reports. In particular, it should include the title and three sections of problem descriptions, implementation and results, and discussions.

(1) Title of the lab, e.g. "Lab 1. Introduction to Artificial Intelligence"
(2) Problem descriptions: how you define the problem
(3) Implementation and results: your code and results
(4) Discussions: how you analyse the results, evaluate the performance, etc.

1.1.5 Save your work

Normally your work would be automatically saved at Google Drive – another advantage of the Colab because you don't need to worry about losing your work accidentally.

Of course you can save a local copy by clicking on “File – Download - Download ipynb”.

You can also save your report to other format, such as PDF. There are a number of ways to convert the .ipynb files into PDF files. Perhaps the most straightforward one is that you do “File – Print – Save as PDF” within Colab.

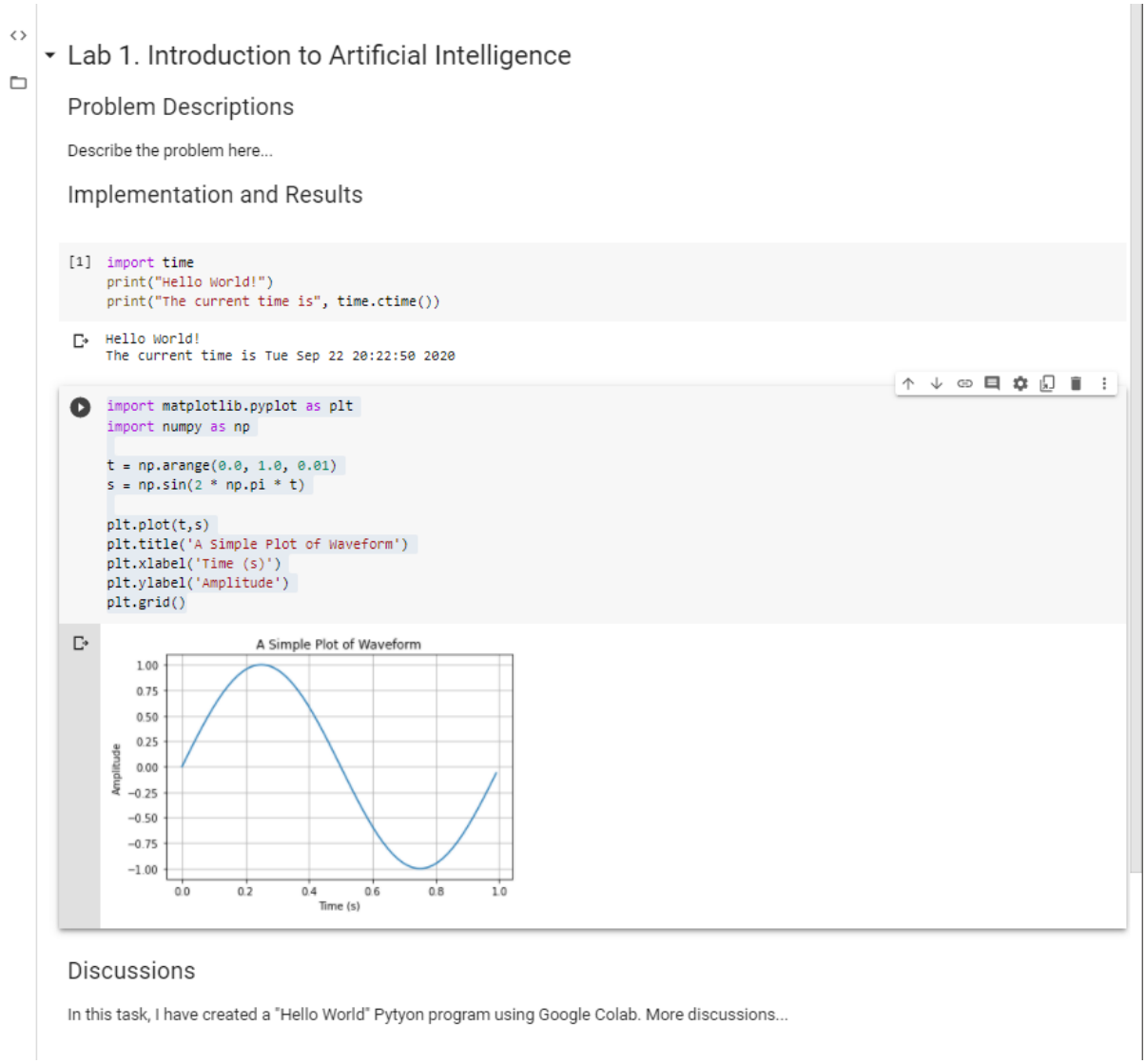


Fig 1.2 Example of a lab report with a title and three sections of (1) problem descriptions, (2) implementation and results, and (3) discussions.

Task 1.2 Python Warm-Up

Without any doubt mastering the Python language is very helpful to you if you plan to pursue a career in Artificial Intelligence and Data Science. Actually one may even argue this is a “must” requirement. However, we understand some of you are unfamiliar with Python, or have little experience with it before. Within the MSc programme, we offer an additional course to help you learn Python quickly.

At the same time, as well as the traditional textbooks on Python programming, there are many online tutorials, resources or even videos available. If you prefer to learn Python interactively with a video, a simple search at YouTube with keywords such as Python, learning beginners etc would direct you with plenty of resources. Normally those with high number of views would be reasonable choices. I would like to recommend the following one. The whole video is about 4.5 hours if play at standard 1x speed and it provides a detailed list of contents where you can easily jump to any topic you are interested.

Learn Python - Full Course for Beginners [Tutorial]
https://www.youtube.com/watch?v=rfscVS0vtbw